PROJECT PROPOSAL

SynthLens: Detecting Synthetic Product Reviews

A Hybrid Approach Combining NLP Vectorization and Stylistic Feature Fusion using Scikit-Learn Pipelines

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1. Proposed Project Topic

SynthLens: Detecting Synthetic Product Reviews.

A Hybrid Approach Combining NLP Vectorization and Stylistic Feature Fusion using Scikit-Learn Pipelines

2. Identification of problem

While exploring real-world problems that can be investigated using the approach of data science and machine learning techniques, and fraudulent activity occurring while navigating online e-commerce platforms like Amazon, AliExpress, Temu, and so on.

With the rapid increase and widespread accessibility of Large Language Models (LLMs), e-commerce platforms, product reviews, and digital marketplaces face an escalating threat of automated review manipulation. Synthetic and AI-generated reviews artificially inflate or deflate product ratings, misleading consumers, undermining fair market competition, and
eroding organizational trust in digital feedback systems.

So, for this problem, we will use modern digital trust and safety research approaches alongside open-source datasets available on public benchmark repositories such as Kaggle (e.g., the DAIGT V2 Train Dataset and LLM - Detect AI Generated Text benchmarks). Traditional spam filters rely on simple keyword matching or metadata anomalies, which fail against natural-sounding LLM generated text. Detecting synthetic reviews requires a data science approach that evaluates fine-grained stylistic patterns and lexical vector representations.

3. Problem to Be Investigated

This project investigates whether natural-sounding synthetic product reviews from AI can be reliably distinguished from genuine human-written reviews using standard machine learning classification models. The primary research question is:

- How accurately can a hybrid machine learning pipeline—combining TF-IDF n-gram vectorization with engineered stylistic features—classify synthetic (AI-generated) vs. human-written customer reviews while preventing data leakage during evaluation?

To keep the project scope well-defined, reproducible, and computationally efficient for the course project, our analysis will focus on English-language text reviews and benchmark supervised algorithms (such as Logistic Regression, Linear Support Vector Machines, and Random Forests) encapsulated inside scikit-learn pipelines.

3. Project Scope & Necessity

The necessity of this project arises from a critical vulnerability in modern digital platforms: legacy fake review detection systems were constructed around simple rules, rate limits, or static keyword lists. Modern LLMs easily bypass these filters by producing grammatically correct, emotionally persuasive text that closely mimics human feedback. Consequently, automated trust verification systems must evolve beyond simple term frequency models to capture both semantic structure and stylistic text fingerprints.

To address this necessity, the scope of this project focuses on three key methodological
objectives:

- Hybrid Feature Space Exploration: While traditional NLP models evaluate either raw text vectors (TF-IDF) or surface-level stylistic statistics separately, this project investigates a hybrid approach. We extract character-level and word-level n-grams while simultaneously engineering fine-grained stylistic features (e.g., character/word count ratios, capitalization variance, punctuation density, and vocabulary richness metrics) to capture subtle stylistic footprints left by synthetic text generators.
- Leakage-Free Pipeline Architecture: A common flaw in applied text classification research is data leakage during feature extraction and pre-scaling. By strictly encapsulating TfidfVectorizer, StandardScaler, and classifiers within scikit-learn Pipeline and ColumnTransformer structures, we ensure that all feature transformations are computed strictly within training folds during cross-validation.
- Interpretable and Reproducible ML Baselines: Rather than utilizing computationally intensive deep neural networks, this project investigates whether fast, transparent, and reproducible linear and tree-based classifiers (Logistic Regression, LinearSVC, Random Forests) can achieve competitive diagnostic accuracy while offering direct feature importance interpretability.

This research scope aligns directly with recent literature in text forensics and machine learning applications (e.g., Crawford et al., 2015; Pedregosa et al., 2011; Müller & Guido, 2016), demonstrating that structured feature fusion and leakage-free cross-validation are essential for reliable digital content verification.

5. Proposed Dataset

The project will utilize open-source benchmark datasets hosted on Kaggle and public repositories, specifically curated for AI vs. Human text detection (such as the DAIGT V2 / Human vs. AI Generated Reviews datasets). The aggregated dataset consists of labeled textual reviews containing:

- Target Variable: Binary indicator label (0 = Human-Written Review, 1 = Synthetic / AI-Generated Review).
- Text Input: Raw customer review text content across various product categories.
- Data Integrity: The dataset will be inspected and preprocessed using pandas to handle null entries, text formatting, and class distribution balancing.

6. Proposed Approach & Methodology

Our implementation will strictly follow data science practices using Python, pandas, and scikit-learn:

- Data Ingestion & Cleaning: Import raw review data into pandas DataFrames, clean text formatting, and audit missing or duplicate entries.
- Stylistic Feature Engineering: Compute review-level numerical metrics, including character count, word count, average word length, capitalization ratio, exclamation mark frequency, and punctuation density.
- Text Vectorization: Extract lexical and syntactic signals from raw text using scikit-learn's TfidfVectorizer (extracting word and character n-grams).
- Feature Fusion via ColumnTransformer: Fuse sparse text vectors and scaled numerical stylistic features into a unified feature matrix using ColumnTransformer.

- Pipeline Construction: Encapsulate feature extraction, numerical scaling (StandardScaler), and classifiers (LogisticRegression, LinearSVC, RandomForestClassifier) inside scikit-learn Pipeline objects to ensure complete isolation and prevent data leakage.
- Hyperparameter Tuning & Cross-Validation: Optimize n-gram ranges, regularization terms, and tree parameters using GridSearchCV integrated with Stratified K-Fold cross-validation.
- Performance Evaluation: Evaluate models using Classification Metrics including Accuracy, Precision, Recall, Macro/Weighted F1-Score, Confusion Matrices, and ROC-AUC curves.

7. References

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